

BREATHEBOX · DESIGN PRECIS

BreatheBox design precis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

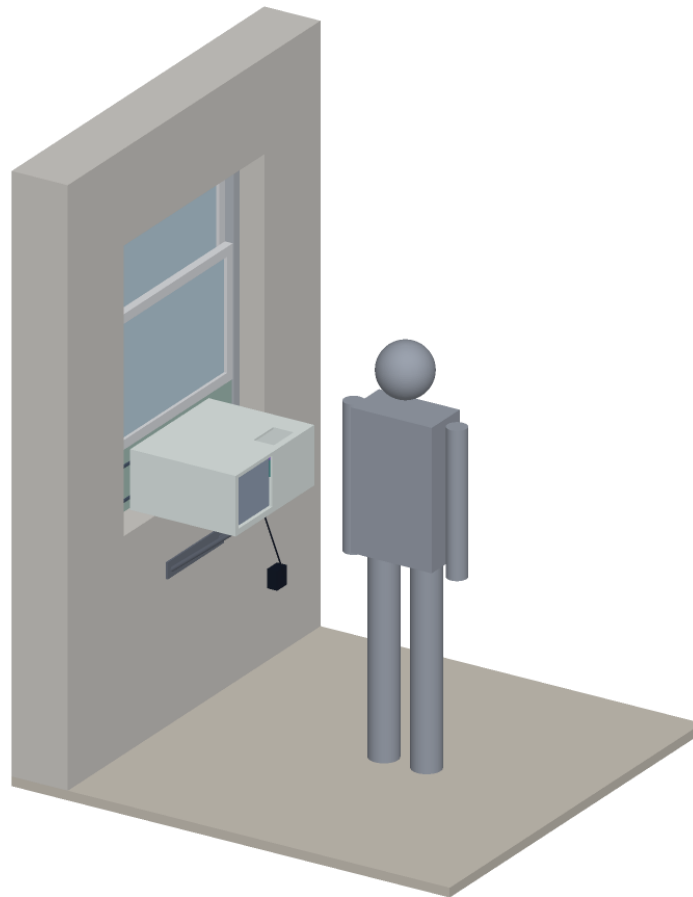
BreatheBox design precis

■ Summary

BreatheBox is a single-room heat recovery ventilator that sits on the sill of a sash window. Two small 24 V fans push stale room air out and draw fresh air in through a counterflow plate core, so the incoming air picks up about 80 % of the heat (or, in summer, the cool) of the outgoing air. First-order estimates for a bedroom at 50 m³/h and 0 °C outdoors: about 270 W of heat kept in the room for about 9.5 W of fan power, CO₂ near 1,000 ppm with two sleepers, and about \$243 in parts, which is over the \$220 budget. All figures are estimates to be checked at TRL 3.

BreatheBox

CONCEPT, NOT FOR FABRICATION



Grey figure: 1.75 m person for scale

Figure 1. Massing model on a sash window, with a 1.75 m person for scale.

■ How it works

1. **Mount.** A 20 mm insulated panel fills the gap under the raised lower sash, which closes down onto it and is locked. The housing rests on the sill and on a clamped bracket; nothing is drilled.
2. **Exhaust path.** Room air enters a front grille, passes a coarse filter and one set of channels in the core, and leaves by the exhaust fan through a downward-facing outdoor hood.

3. **Supply path.** Outdoor air enters the second hood, about 420 mm from the exhaust outlet, passes an ePM1 filter and the other set of core channels, and the supply fan blows it into the room through a top grille that throws air up and away from the occupant.
4. **Recover.** The two streams flow in opposite directions on each side of thin polymer plates, so heat passes across without the air mixing.
5. **Drain.** In winter, moisture from the room air condenses in the core. A tray under the core falls to the outdoor side and a tube carries the water outside.
6. **Control.** A small controller reads CO₂, humidity and temperature at the room intake and runs the fans between 30 and 70 m³/h. A temperature probe at the core's exhaust outlet triggers a frost mode that slows the supply fan when the outgoing air nears 0 °C. Readings show on a status LED and, optionally, on a local web page; nothing needs the cloud.

BreatheBox: cutaway

CONCEPT, NOT FOR FABRICATION

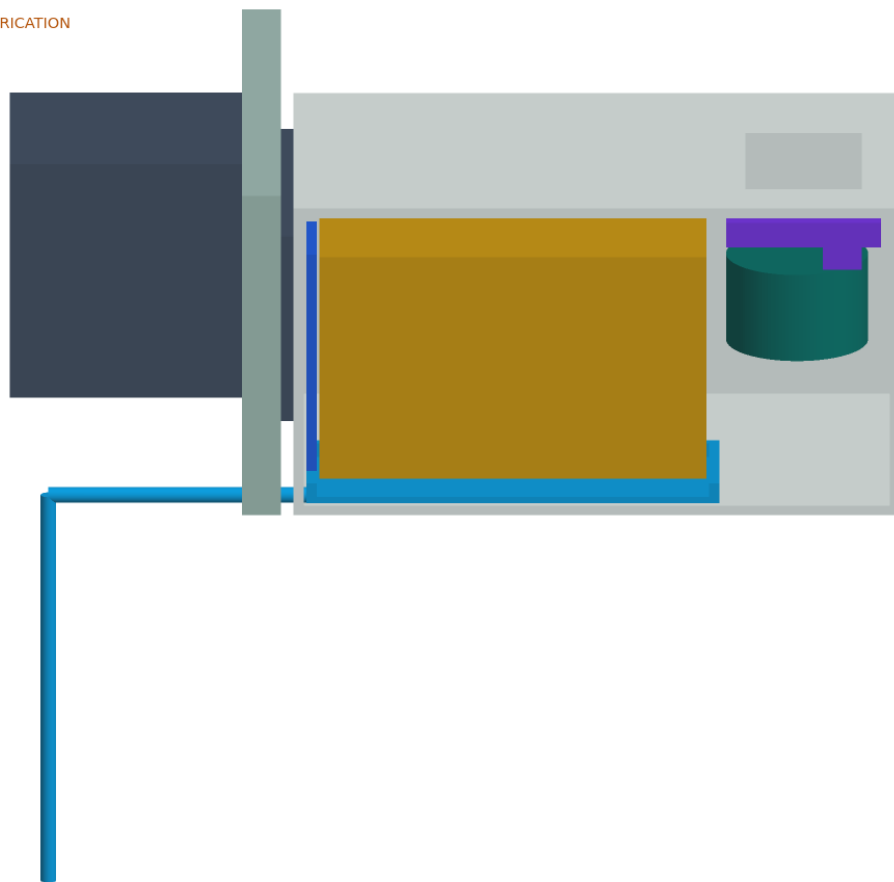


Figure 2. Cutaway through the supply side: hood and duct (outdoors, left), insert panel, filter, core over the condensate tray, supply fan and controller.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 4.

#	Component	Proposed choice	Notes
1	Insulated housing	PVC foam board shell with 10 mm foam lining, about 470 x 560 x 250 mm	Lift-off lid for filters
2	Counterflow core	Polymer plate core about 180 x 180 x 300 mm	Spare-part HRV core; washable. Proposed, awaiting Amish
3, 4	Supply and exhaust fans	24 V brushless centrifugal blowers, 120 mm class, PWM with tach	Tach allows flow balancing
5	Supply filter	ISO 16890 ePM1 50 % panel	Replace about every 6 months
6	Exhaust filter	Coarse washable pad	Keeps lint off the core
7	Controller	ESP32-C3 class module with a Sensirion SCD41 class CO ₂ , humidity and temperature sensor and two NTC probes	Check the sensor on the lab's CalRig before use
8	Condensate tray and drain	PETG tray, 12 mm tube to outdoors	Falls outward
9	Window insert panel	20 mm insulated panel with EPDM edge seals, trimmed to 700 to 1,000 mm	Sash locks onto it
10	Outdoor hoods and ducts	Two PVC hoods with 1 mm stainless mesh	Openings face down
11	Sill bracket	Aluminium plate and struts to a padded wall foot	Clamps; no drilling
12	Power supply	Certified 24 V, 1.5 A plug-in adapter	No mains wiring in the unit

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3. Assumptions: air density 1.2 kg/m³, specific heat 1,005 J/(kg·K), design case as in BBX-REQ-001.

Table 2. First-order estimates.

Quantity	Estimate	Basis	Requirement
Heat capacity rate of the airflow	about 16.8 W/K	50 m ³ /h x 1.2 kg/m ³ x 1,005 J/(kg·K) / 3,600 s/h	
Heat carried by exhaust air at 20 K difference	about 335 W	16.8 W/K x 20 K; this is also the loss from an open window giving the same airflow	
Heat recovered	about 268 W; about 67 W still lost	80 % sensible effectiveness	R2 met on paper
Supply air temperature at 0 °C outdoors	about 16 °C	0 + 0.8 x 20 K	
Fan and control power	about 9.5 W	Two fans at about 4.5 W each (about 1.1 W air power at 80 Pa, about 25 % fan efficiency), controls about 0.5 W	R3 met on paper
Heating-season energy kept	about 1,100 kWh per season	16.8 W/K x 24 h x 3,500 K·d (assumed cold-temperate climate) x 0.8	
Fan energy over the same season	about 50 kWh	9.5 W for 212 days	
CO ₂ , two sleeping adults	about 980 ppm at 50 m ³ /h; about 820 ppm at 70 m ³ /h	Steady state, 0.014 m ³ /h per sleeper, 420 ppm outdoors	R4 marginal at 50, met at 70
Condensate, upper bound	about 0.14 L/h	Room air 7.3 g/kg cooled to about 4 °C (saturation about 5.0 g/kg), 60 kg/h of air	R7 drain sized for 0.15 L/h
Frost onset	around -5 °C outdoors	Exhaust leaves the core at about 20 - 0.8 x (20 - T _{out}); at -5 °C this is about 0 °C	R7 at risk
Clear distance between hood openings	about 420 mm	Massing model	R11 met on paper
Unit mass	about 10 kg	Housing 3 kg, core 1.5 kg, fans 0.8 kg, panel 1.5 kg, hoods 1 kg, bracket 1.5 kg, others 0.7 kg	R9 met on paper
Parts cost	about \$243	Indicative prices, bom/bom.csv	R15 not met (\$220 target)

BreatheBox: heat flow at 0 °C outdoor, 20 °C indoor, 50 m³/h (estimates)

CONCEPT, NOT FOR FABRICATION

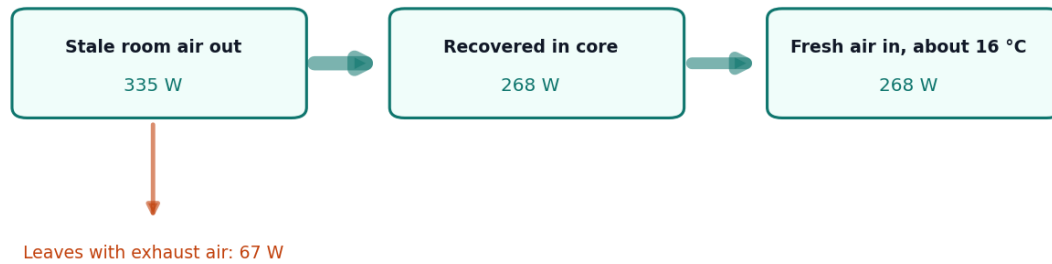


Figure 3. Heat flow in the design case (estimates).

■ Key design choices

Each choice below is proposed, awaiting Amish.

- **Window insert rather than wall core.** Renters can install it, and it comes out without a trace. The cost is that it only suits windows with a vertical or horizontal sliding sash (R8).
- **Continuous counterflow core rather than a pair of alternating regenerative units.** A plate core runs one steady fan per stream, is simple to model and build, and keeps supply and exhaust separate. Alternating ceramic units avoid condensate drainage and frost more gracefully but need two wall openings and paired, reversing fans.
- **Buy the core, make the rest.** The core is the part a garage builder cannot make well. A home-made crossflow core from corrugated plastic would cost about \$10 but recover only about 50 to 65 % (estimate).
- **Frost by slowing the supply fan.** The simplest frost strategy. It unbalances the flows for short periods and slightly depressurizes the room (see Safety). A small preheater would keep balance but adds mains-level power and cost.
- **24 V SELV only.** A certified plug-in adapter means no mains wiring for the builder (R10).
- **Sensible-only core.** A cheaper, washable polymer core. An enthalpy (moisture-transfer) core would suit hot, humid climates and dry winters better, at a higher price.

BreatheBox: exploded view

CONCEPT, NOT FOR FABRICATION

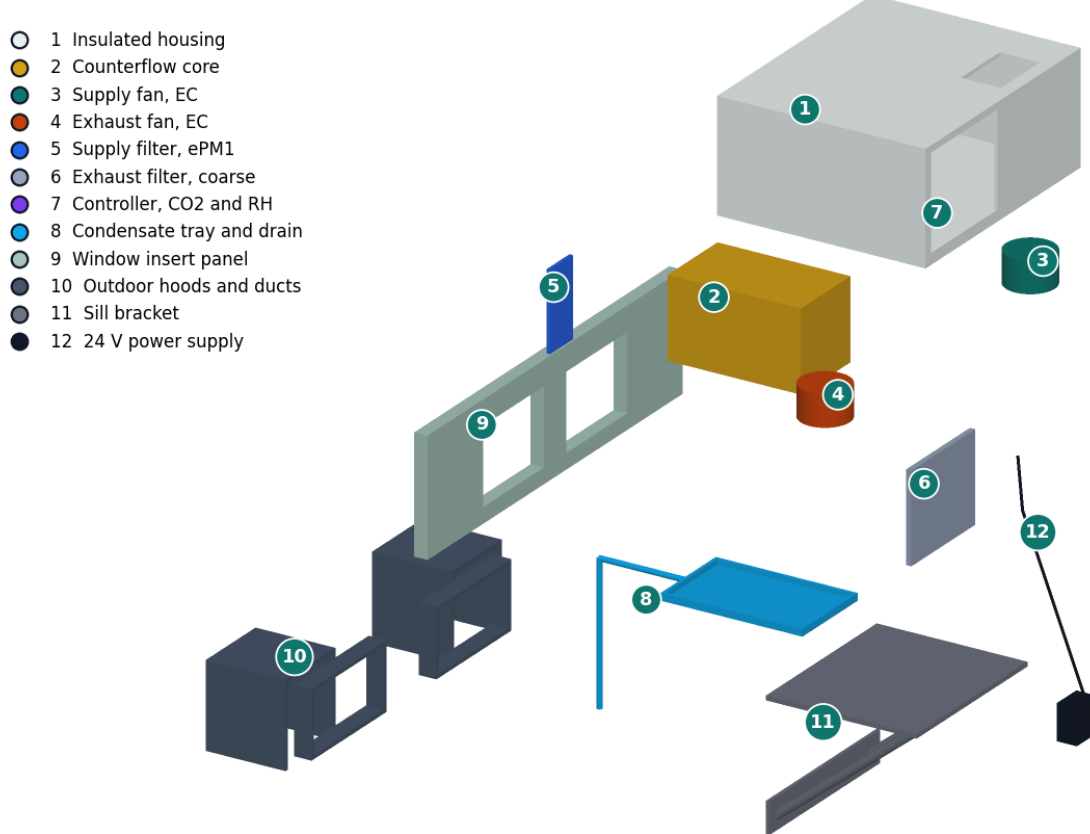


Figure 4. Exploded view with BOM numbers.

■ Safety

Safety: Do not use BreatheBox in a room with an open-flued or unflued fuel-burning appliance (gas fire, oil or wood stove, water heater). In frost mode the unit extracts more air than it supplies and could draw combustion gases into the room.

Safety: The raised sash must be locked onto the insert panel. An unlocked sash is a security risk, and a loose panel could fall outward from an upper floor. Check the bracket and panel before leaving the unit unattended, and never install from outside above ground level.

Safety: Use only a certified 24 V SELV plug-in adapter, with a fuse on the low-voltage input. Do not open or modify the adapter. Keep the cable clear of the sash and of the condensate tube.

- Foam and plastics in the housing burn; choose fire-retardant grades and keep the unit away from heaters and curtains near heat sources.
- Fan impellers can cut fingers: guards or grilles on both room openings, and disconnect power before opening the lid.

- Clean the tray and replace filters on schedule; standing water and wet filters can grow mould.
- The CO2 reading is a ventilation indicator only and is not a health or medical measurement.

■ Open questions for TRL 3

- Fan operating point: do 120 mm class blowers deliver 50 to 70 m³/h against the estimated 80 Pa, quietly enough for R6?
- Frost: how long does frost mode run at -10 °C, and what is the net depressurization?
- Core: which spare-part cores are available, at what size, effectiveness and pressure drop?
- Airtightness: how much air leaks around the insert panel and sash, and does it matter for balance?
- Window survey: what share of the first users' windows are sash, slider, casement or tilt-and-turn?
- Controller: reuse a common lab controller, or a single module as proposed?

Concept media: [blueprint sheet](#), [interactive 3D model](#).